

Blood Pressure Report Guidelines

Introduction

These are some prompts and suggestions to guide you in writing a formal lab report. For example, for this weeks lab, for the introduction write 1-3 paragraphs that covers the following: Explain why blood pressure is important, and why it would be advantageous to regulate blood pressure for certain situations. Define systolic and diastolic blood pressure, and how blood pressure varies from the central circulation (heart) to the periphery. Introduce the dive response and end with a statement of your hypothesis.

Methods

Briefly describe what parameters you measured in this experiment (heart rate, blood flow to the periphery, leg volume, etc.). Explain how you will measure these parameters, describe the subjects, and what the experimental treatment(s) were. Follow the lab report guidelines, include all relevant components. Remember not to write step-by-step procedures bur rather communicate the experimental desing with enough detail to repeat the experiment. Refer to standard techniques in your lab manual and cite. Briefly describe how you will analyze the data.

Data Collection (for your notebook)

In your lab notebook, collect the following data (recreate a table like the one below) with your group's data from this experiment (one column/individual). You will need this information to write a lab report.

Measurement	Systolic Pressures				Diastolic Pressure			
Upper arm								
Forearm								
Leg (lying down)								
Leg (standing)								

Questions for Thought

Before you go home, write answers to the following questions in complete sentences in your notebook.

- 1) What do the Korotkoff sounds represent?

- 2) Explain the events occurring in the heart during: (a) systole, (b) diastole.
- 3) How does your estimate of systolic pressure in the upper arm measured by auscultation compare with your estimate using computer data recording?
- 4) Does systolic pressure differ between the forearm and upper arm?
- 5) How does your estimate of systolic pressure in the leg compare with that from the upper arm?
- 6) What happened to blood pressure in the leg when you were standing up?

More Specific Guidance for the Dive Response

Results

The result section must include explanatory text in paragraph form as well as figures and tables that support the written results. Figures or tables should make an important point (i.e. the treatment had an effect or did not have an effect), and should not be redundant. For example, you should not have both a figure and a table of the same data. For each figure or table that you include, briefly explain in text what that figure or table shows (1-2 sentences is good enough in most cases). See format instructions.

In general, do not present raw data in a formal lab report, instead present analyzed data such as means. Think about what form of the results best addresses your hypothesis. For example you may include mean systolic and diastolic pressures before and after a treatment, if a change in mean addresses your hypothesis the best. Make sure to include standard deviations or standard errors if you are presenting means. The only exception in physiology is that we do often include one example of a good waveform (for example of a pulse) to show that we are collecting good data.

The following are a list of potential waveforms and graphs that may help you write a convincing lab report. Be sure your results are written to illustrate your thought process in identifying the factors which the dive response.

The Pulse Waveform

Include pictures of the Zoom Window below showing your pulse beat waveform during rest, breath-holding, and during a dive.

Leg volume

Include a representative picture from the Chart Zoom Window of your leg volume data tracing during rest.

Effects of breath-holding on heart rate

Make a bar graph showing your heart rate (BPM) during the following conditions: Rest, Breath hold, and Recovery.

Effects of breath-holding on peripheral circulation

Make a bar graph showing your relative leg volume (% of resting value) during the following conditions: Rest, Breath hold, and Recovery.

Effect of diving on heart rate

Make a bar graph showing your heart rate (BPM) during the following conditions: Rest, Dive, and Recovery.

Effect of diving on peripheral circulation

Make a bar graph of relative leg volume (% of resting value) for each of the following conditions: Rest, Dive, and Recovery.

Additional Experiments

Graphically summarize the results of the additional experiments that your group conducted, following the format of the above graphs.

Conclusions

These are suggestions for your discussion. The discussion must flow naturally, but do work in answers to the following questions in complete sentences in order to add intellectual content:

- 1) Did the amplitude/stroke volume of your pulse beat change when you held your breath? During a dive? Explain.
- 2) How did your leg volume change during breath holding? During a dive? What do your results suggest about the rate of blood flow to your extremities during diving?
- 3) Did you observe bradychardia during a breath hold? How about a dive?
- 4) What was the time course of the diving response, and how coordinated was it? How long did the response take to begin, and how quickly did your heart rate return to normal after a dive?
- 5) What was the average change in heart rate during a dive for your class/group? Was the change similar in everyone? The diving response is often considered to be adaptive. Do you think this is so? Explain.